

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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CLI To-Do List Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

This project defines a data model for a CLI To-Do List Manager utilizing a local JSON file (`~/ .todos .json`) for persistence. It implements a sequential ID-based task system with strict validation on field types and state transitions (pending, completed, deleted). The architecture focuses on data integrity, ensuring immutability of creation timestamps and strict sequential ID generation.

1.2 Maturity Assessment

The specification provides a solid foundation for the data schema but is currently in REFINEMENT. While the field definitions are precise, there is a critical lack of formal Acceptance Criteria and a Testing & Validation strategy, particularly regarding boundary values and corruption recovery. The absence of defined external dependencies for ISO 8601 validation also requires clarification before full implementation.

1.3 Technical Stack

- **Storage Format:** JSON
- **Timestamp Standard:** ISO 8601 UTC
- **Persistence Layer:** Local Filesystem (`~/ .todos .json`)

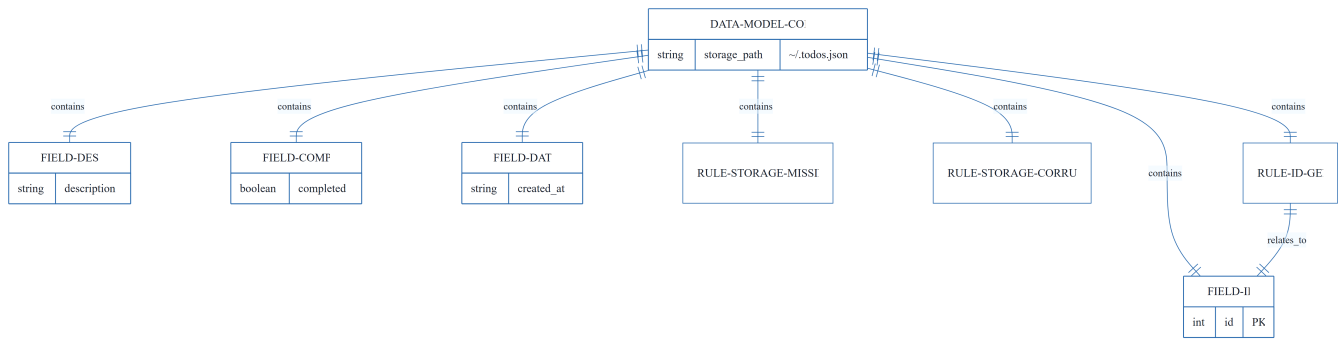
1.4 Architectural Constraints

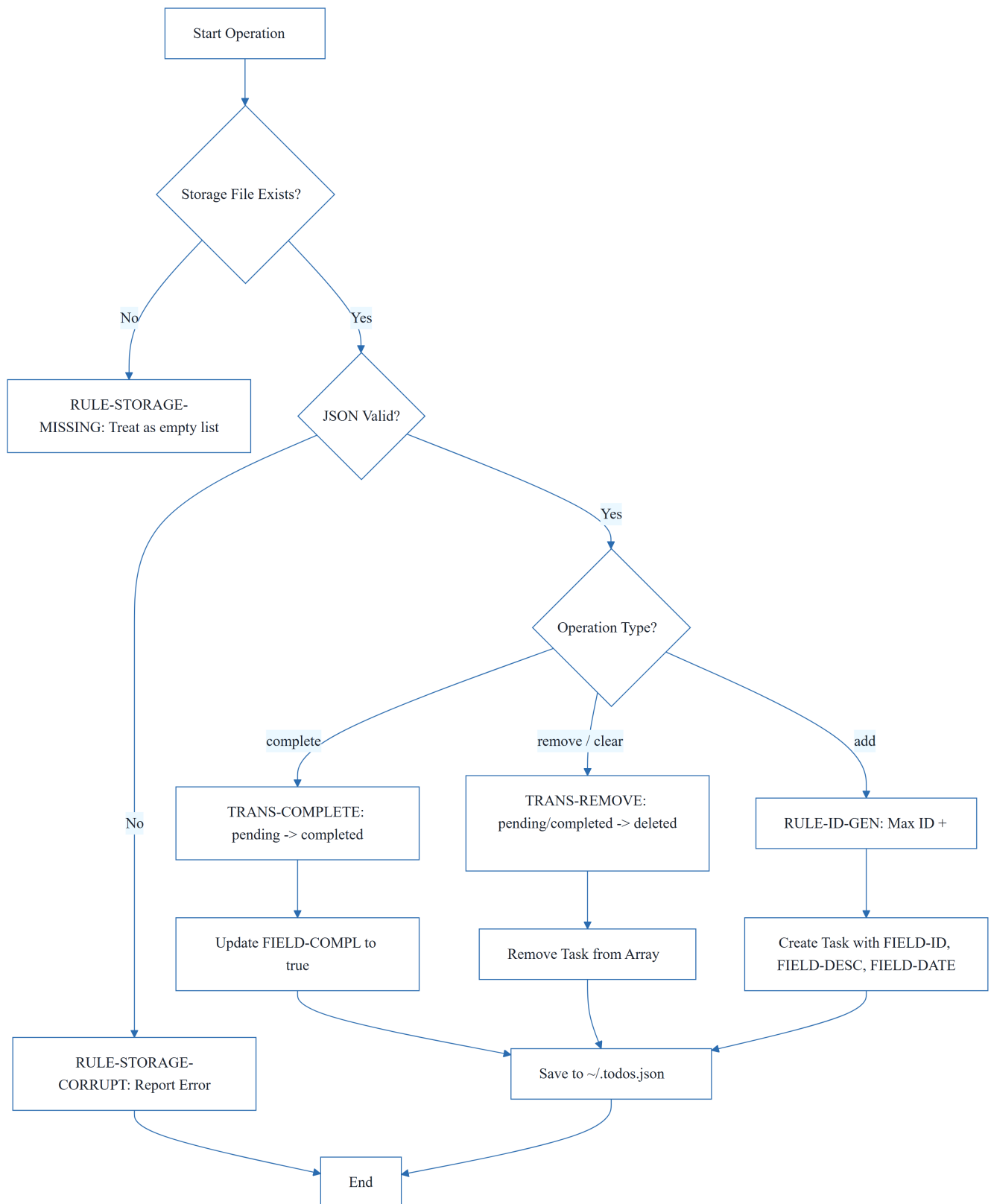
- **ID Generation:** IDs must be strictly > 0 , unique, and sequential (Highest current ID + 1).
- **Data Integrity:** `created_at` timestamps must be immutable after the initial creation.
- **Storage Fallback:** A missing storage file must be interpreted as an empty task list.
- **Error Handling:** Corrupted JSON files must be reported as a user error and halt operation.
- **Field Validation:** Descriptions must be trimmed and contain non-whitespace content.

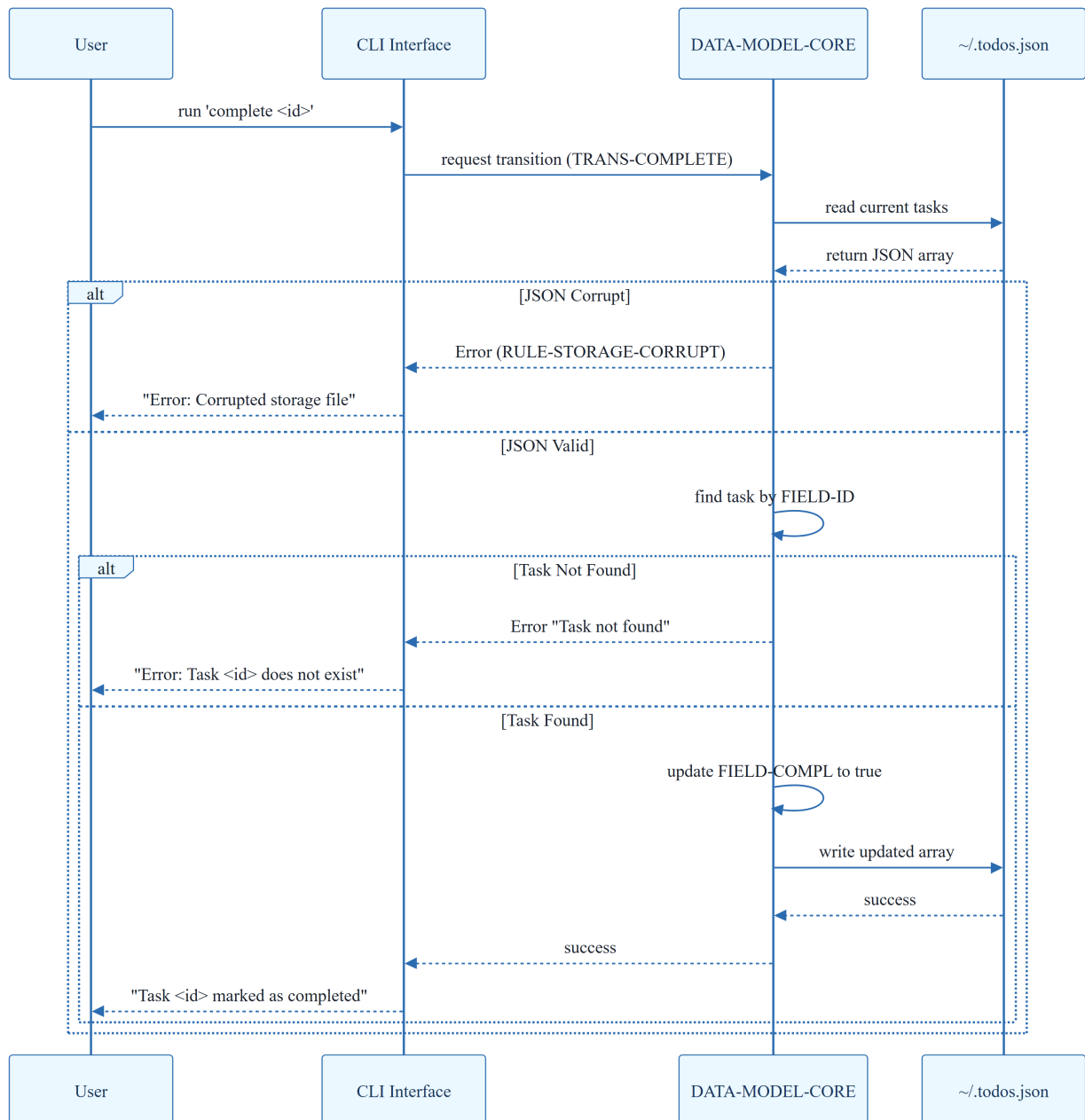
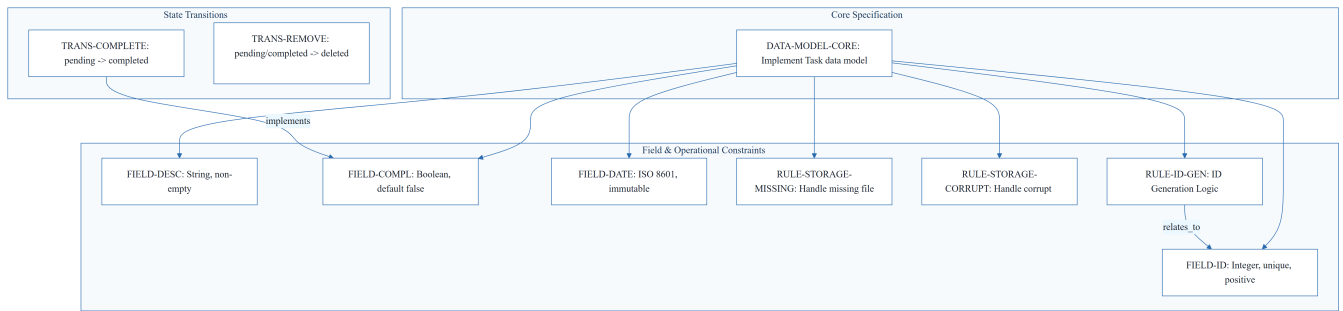
1.5 Critical Dependencies

- Local filesystem access for `~/ .todos .json`.
- JSON parsing engine for data serialization/deserialization.
- ISO 8601 compliant timestamp generator.
- Referential integrity between Task ID and state transition commands (`complete/remove`).

2. Architecture Workflows

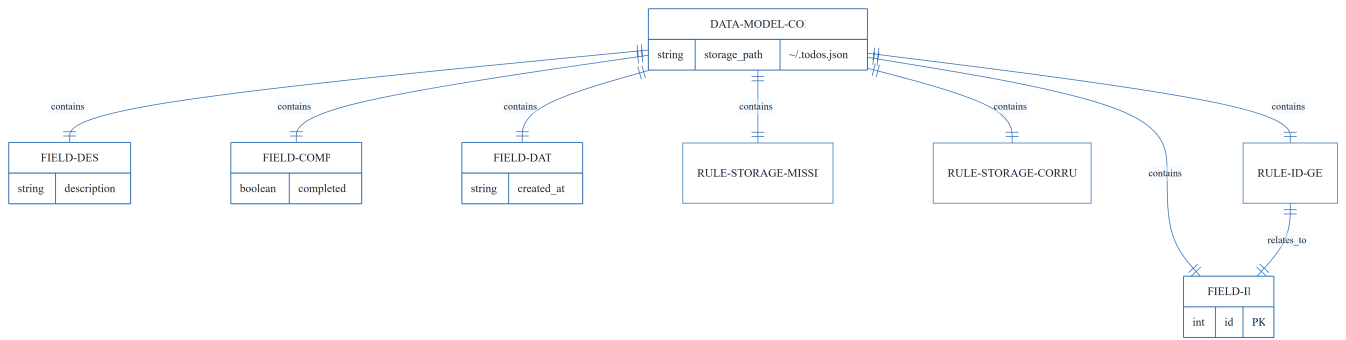




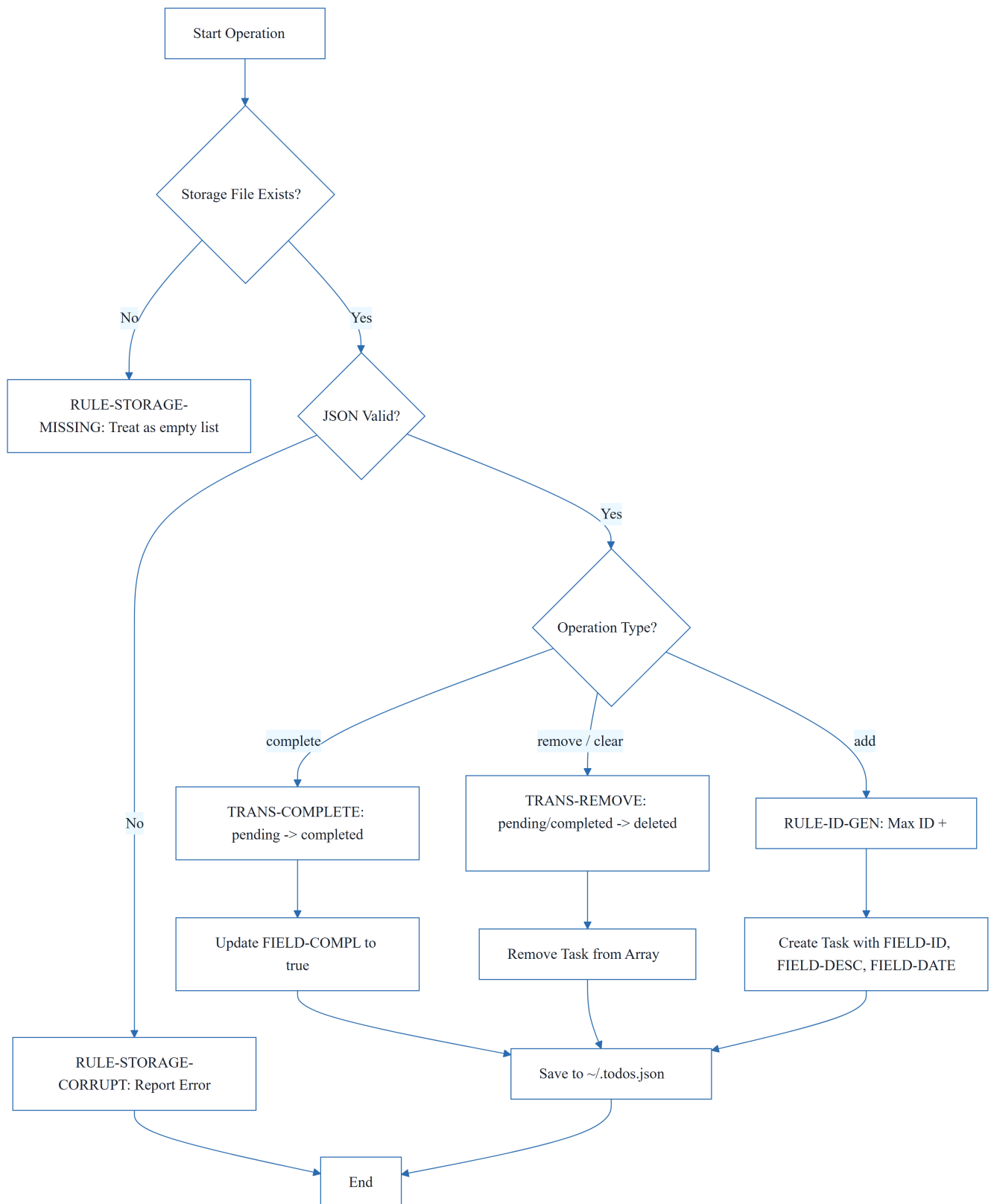


& Visual Diagrams

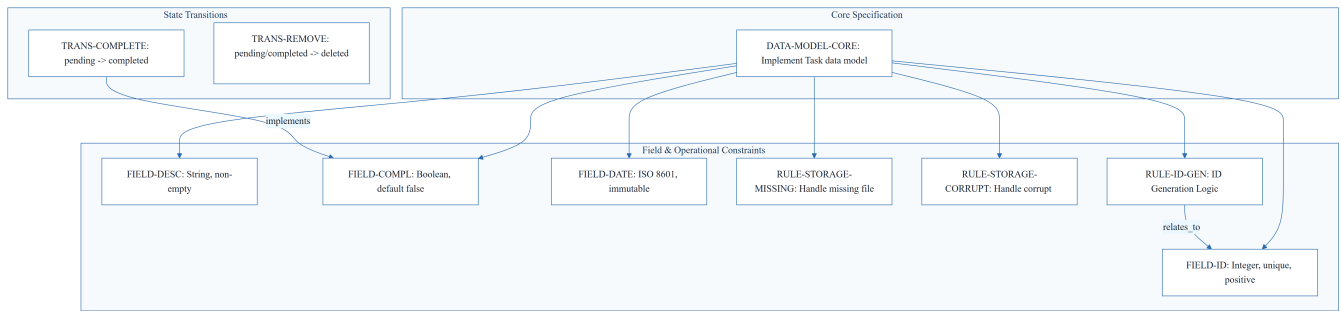
2.1 CLI To-Do Manager Data Model



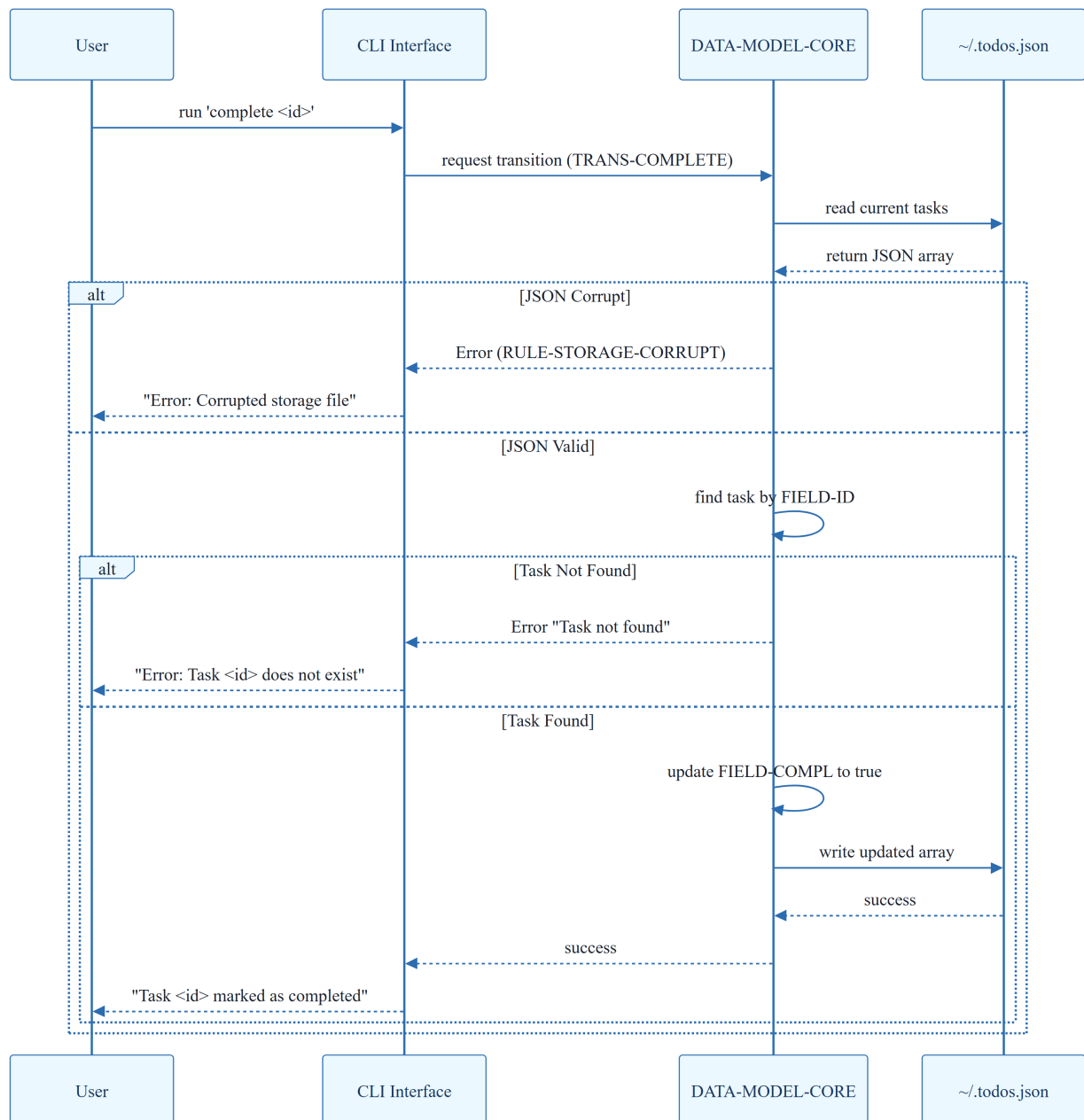
2.2 Task State Transition Workflow



2.3 Data Model Traceability Map



2.4 Task Lifecycle Sequence



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section
DATA-MODEL-CORE	Task	Implement the Task data model stored in ~/.todos.json	Task
FIELD-ID	Constraint	id: Integer, unique, sequential, positive, required (> 0, no duplicates)	Fields
FIELD-DESC	Constraint	description: String, required, trimmed, non-empty	Fields
FIELD-COMPL	Constraint	completed: Boolean, required, defaults to false	Fields
FIELD-DATE	Constraint	created_at: ISO 8601 UTC timestamp, required, immutable	Fields
RULE-STORAGE-MISSING	Constraint	Missing storage file must be treated as an empty task list	Operational Rules
RULE-STORAGE-CORRUPT	Constraint	Corrupted JSON must be reported as an error to the user	Operational Rules
RULE-ID-GEN	Constraint	IDs assigned from highest current ID + 1	Operational Rules
TRANS-COMPLETE	Sub-task	Transition pending -> completed via 'complete '	State Transitions
TRANS-REMOVE	Sub-task	Transition pending/completed -> deleted via 'remove ' or 'clear'	State Transitions

3.2 Security Rules

- **Data Integrity:** The `created_at` field must be immutable. Any update operation must preserve the original timestamp.
- **Input Validation:** The `description` field must be trimmed of whitespace; empty strings or strings containing only whitespace are prohibited.
- **Type Safety:** Strict enforcement of types: `id` (Integer), `description` (String), `completed` (Boolean), `created_at` (String/ISO 8601).

3.3 Data Models

Storage Entity: Task

- **Storage Path:** `~/todos.json`
- **Structure:** JSON Array of Objects.

Field Definitions:

- `id`: Primary Key. Positive integer.
- `description`: Task content. Non-empty string.
- `completed`: Status flag. Boolean.
- `created_at`: Creation date. ISO 8601 UTC string.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Dependencies & Integration Points	LOW	Define if any external libraries are needed for JSON parsing or ISO 8601 validation.
Acceptance Criteria	HIGH	Define specific testable outcomes for the data model (e.g., 'Verify that a task cannot be created without a description').
Checkboxes Checklist	MEDIUM	Create a checklist for implementation (e.g., [] Implement JSON read/write, [] Implement ID increment logic).
Testing & Validation	HIGH	Define test cases for corrupted JSON files and boundary values for IDs.
Open Questions & Uncertainties	LOW	Check if there are limits on the number of tasks or file size.

4.2 Remediation & Workflow

The project is currently in the **Refinement** phase. The immediate priority is to address the **HIGH** priority gaps by defining a formal Testing & Validation suite and a set of Acceptance Criteria to ensure the data model behaves as expected under edge cases (e.g., file corruption, ID collisions).

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
ID	BUSINESS_DOMAIN	FIELD-ID	A positive, sequential, and unique integer used as the primary key for each entry, calculated by incrementing the maximum existing value by one.
JSON	TECHNICAL_STACK	RULE-STORAGE-CORRUPT	The lightweight data-interchange format used for the persistent storage file located at ~/.todos.json.
UTC	TECHNICAL_STACK	FIELD-DATE	The time standard used for the immutable creation timestamp, formatted according to ISO 8601.
complete \	BUSINESS_DOMAIN	TRANS-COMPLETE	The operational command that triggers a state transition from pending to a finished status for a specific record.
remove \	BUSINESS_DOMAIN	TRANS-REMOVE	The operational command that triggers a state transition from any active status to a deleted state for a specific record.